

Research ASSIGNMENT 1

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Course: Data Analytics

Section A: Data Fundamentals

1. A database is a structured system where we store and organize data making / simplifying things for us to retrieve, manage and analyze information [data]. Relational data
In the modern world transaction processes like banking or paying in store using bank cards it is a form of data being captured and retrieved also personalized user experiences like social media.

2. Data Warehouse

- It is a central source where multiple databases are stored, it houses large amounts of data from wide range of sources including transactional systems, self service analytics and customer related feedback which in return gives powerful insights to businesses to explore and analyze for affirmative action ~~that~~ within the organization and support better decision making.

3 Data Lake

- It stores and processes data but the data stored is raw, semi-structured and unstructured from sources such as mobile apps, IoT devices and social media using low-cost cloud object storage.

within a single platform, data warehouses caters for structured data which is in a form of relational data immediately ready for analysis. businesses would rather use data lakes for multipurpose data storage, data they are not ready to use yet, and also archive old data.

4. Data Mart

- Because a data warehouse contains data from different areas and positions within an organization which can be confusing and overwhelming when fishing for specific data, data mart makes things easy as it ring-fences certain data sources for certain people within the organization whom are allowed access to or to work with, data security and compliance is promoted by data mart

Marketing		Sales	
☐	☐	☐	☐
☐	☐	☐	☐
☐	☐	☐	☐
☐	☐	☐	☐
Finance		Operations	

the above illustration indicates that the marketing team ~~will~~ within the organization will have access to the marketing data mart; sales to the sales data mart and so on. data mart increases team efficiency, reduce costs.

5. a) structured data [rows & columns are related to each other]
- data stored in a form of rows and columns which is more organized, clean and easy to interpret as it -

has been cleaned and summarised ~~for~~ through Analysis to make insightful decision for a business
e.g. Relational database; spreadsheet; excel files; Application forms etc.

b) Semi-structured data

A combination of structured and un-structured data for example emails

c) un-structured data

- Data is irregular as it has no pre-defined format and structure and it is ambiguous because it is uncertain in nature meaning it can contain anything from text, images or files, videos; Audios.

6. Data Modeling

- A visual blueprint of data showing the connection between data and its relationships, and the types of data used and stored in the system so it is easy to share the data used
- Conceptual data modeling → A high-level, visual representation of business data which shows the relationship between the business entities such as customers and orders; Product
- Logical data modeling → A detailed blueprint that describes the data elements and relationships within a specific organization without relying on specific database technology.

7. Data Mining

- It is a process of sifting through large data sets to -
Identify patterns and relationships that can help the business
solve problems through data analysis and the use of
machine learning. It helps with marketing, fraud
detection and Improve customer service

8. Data Management System

Technology or source used to organize and store various types
of data. Making sure it is readily accessible when required.

- ① MySQL → used for Social Networking and community sites
e.g. facebook, LinkedIn use MySQL to store user profiles.
- ② Oracle Database → commonly used for OLTP, where data
consistency is vital like banks and e-commerce
platforms [High speed transaction Environments]
- ③ MongoDB → commonly used to store diverse content types e.g.
videos, articles, images]
- ④ Postgres → commonly used in financial system / online Transaction
Processing [OLTP]

9. ETL [Extract, Transform, Load]

→ It is a critical step for data processing as it retrieves raw
data from various sources, cleans and converts it into structured data
→ ETL is vital for data analytics because it combines Big data,
data from various sources into one simplified Information / Table

- ① Extract → Raw data is exported from various sources
structured or un-structured
- ② Transform → here data undergoes a process of cleansing,
Aggregating and putting data in Tables
- ③ Load → data is moved from staging area to target datawarehouse

10. Data Storage formats

① CSV (Comma-Separated values)

→ Text based tabular data, It is simple and commonly used for exporting spreadsheets or databases, CSV are not suitable for large data sets

② JSON (Javascript Object Notation)

→ Text based data format for storing and sharing data across web applications. good for storing structured data

③ Parquet file format - columnar format

→ used for large data sets and real time big data processing
→ Not text-based, humans can't read and open this file format

④ Avro

→ used in big data environments for fast data storage and retrieval

Section B: AI & machine learning

21. Technology that enables computers and machines to simulate human learning, comprehension, problem solving, in ways that used to require human intelligence.

① Narrow AI → designed to perform a single specific task like face recognition or voice assistance

② General AI → hypothetical advanced form of AI that can replace a human being

12. Machine Learning

- supervised learning: Trains models on labeled data predict new, ~~new~~ unseen data.
- unsupervised learning: Finds patterns or groups in unlabeled data
- Reinforcement learning: Learns through trial and error to maximize rewards; ideal for decision making tasks
- So basically this subsets of AI enables systems to learn from data, identify patterns and make decisions without explicit hard-coded instructions / programmed.

13. Deep learning

- A subset of machine learning, driven by multilayered neural networks designed to look like the human brain, computers are enabled to uncover patterns and make informed decisions from vast amounts of unstructured data.

14. Natural Language Processing

- helps convert human language into meaningful data and can also generate human-like language; here humans interact with computer as computers can develop models that can analyse linguistic structure; NLP applications are voice assistants, text analysis, chatbots, content recommendation etc.

15. What is Large language Model (LLM)

- Advanced AI systems built on deep neural networks designed to process understand and generate human-like text
- ① LLaMA 3 (Meta): popular in research and startups
- ② Gemini 1.5 (Google Deepmind): long-context reasoning
- LLM and SLM differ in their size; intended use case, depth of performance; speed is not the same.

16. Generative AI

- The use of AI to create new content, like text, images, music, audio and videos.
- This tool uses ML model to learn and identify patterns and relationships in a dataset of human-created content. It then uses the learned patterns to generate new content.
- Chat GPT - best for text generation, coding and research
- GitHub → best for AI-assisted coding
- Midjourney - best for image generation.

17. Training Dataset

- A machine learning model is taught how to find patterns and truly learn to make predictions on its own. Models with large and good quality training data usually perform better.
- Data bias might lead to unfairly represented data or otherwise discriminate against certain groups or individuals.

18. Computer Vision

- It is a field of Artificial Intelligence that enables machines to ~~interp~~ interpret and understand visual information from images and videos. It uses image processing techniques and deep learning models to detect objects and form visual data like face recognition or video motion analysis.

Section C: APIs, Tools & Developer Concepts

19. Application Programming Interface

→ invisible back bone of modern software development. Applications and systems are enabled to communicate and share data efficiently.

20. API Key

→ unique Identifier used to Authenticate software and systems attempting to access other software or systems of Exposed Applications loose security and sensitive data gets exposed

21. REST API [Representational State Transfer]

→ A well known architectural style used for providing standards between computer systems on the web making it easy for systems to communicate with each other.

HTTP methods → Get: retrieves a specific resource (by ID)
Post: creates a new resource
Put: updates a specific resource (by ID)
Delete: Removes a specific resource (by ID)

22. JSON [Java Script Object Notation]

→ light weight format for storing and transporting data often used when data is sent from server to a web page.

```
{  
  "employees": [  
    { "firstName": "John", "lastName": "Doe" },  
    { "firstName": "Jane", "lastName": "Dijane" }  
  ]  
}
```

23. Prompt Engineering

→ plays a vital role in ensuring accurate, relevant and safe interactions amongst LLMs, It is important because it directly influences the quality, relevance and accuracy of Generative AI

Bad prompt: "Design a menu"

Improved prompt: Design a minimalist one page menu for a high end sushi restaurant, use a color palette of deep blue and gold. style should be modern featuring clear - easy to read - font

Bad prompt: "Fix my code."

Improved prompt: "Act as a senior python developer, identifying why the following code is throwing a 'KeyError' and suggest a more robust way to handle lookups"

24. MAP Function

→ A programming concept where a specific operation is applied to every item in a collection (like a list) to create new collection of transformed results.

```
S = ['1', '2', '3', '4']
```

```
res = map(int, S)
```

```
print(list(res))
```

25. Cloud Computing

- Is the delivery of computing services including servers, storage, databases, software, networking over the internet to offer fast innovation.

IaaS (Infrastructure as a Service)

- here you rent IT infrastructure - servers and virtual machines (VMS), storage, networks, OS systems from a cloud provider on pay as you go.

PaaS (Platform as a Service)

→ made to make it easier for developers to quickly create web or mobile apps, without worrying about managing servers, storage, network and databases needed for development.

SaaS (Software as a Service)

- method of delivering software applications over the internet, on demand and typically on a subscription basis.

26. Version Control System

- a software tool used in software development to track, manage, and record changes to source ~~code~~ code over time.

Git - is the most popular version system in the world. It records changes over time made to our code via repository database; it is known for its speed and flexibility.

Branch: A duplicate of a code base used to develop features or fix bugs without affecting the main code.

Commit: A snapshot of changes made to a codebase at a specific time, used to track and manage project history.

27. Jupyter Notebook

→ It is an interactive web application that allows you to write and run code, languages used are Julia, Python & R.

28. Python

- It is a high-level programming language known for its simple and readable syntax.

Python libraries

- Pandas: → Good for data manipulation and analysis.
introduces high performance data structures

- NumPy: Numerical Python is the foundational library for scientific computing. It supports large, multi dimensional arrays and matrices

matplotlib: It is for creating static, animated and interactive visualizations, serves as a primary tool for generating plots, histograms, and bar charts for exploratory data analysis [EDA]

Scikit learn: it offers simple and efficient tools for predictive data analysis, including algorithm for classification, regression, clustering and dimensionality reduction

Section D: Data Ethics, Governance & Careers

29. Data Privacy

- It goes along with POPI act, a principle that a person should have control over their personal data, including having the right to decide which organizations collect, store and use their data
- The GDPR is a European Union law that governs how personal data is/must be handled, organizations within and outside the EU are affected by this law as long as they collect or use personal data of EU residents

30. Data Ethics

- refers to the principles that govern how organizations gather, protect, and use data / as personal data

30. unethical use of data

- Where personal data is collected, analyzed and stored without consent

1. Spread misinformation $\rightarrow$ AI Algorithms can be exploited to generate false news the spread causing political instability and economic disruption.
2. Security risks and financial loss $\rightarrow$ poorly protected data may lead to cases of fraud and identity theft
3. Breaches of trust $\rightarrow$ Organizations that mishandle data face risks of lawsuits, reputational damage and fines also some organizations like your gambling Industry they may take away their ~~license~~ licence of practice.

31. Data Governance

- Principles that must be met by organizations to manage data during its life cycle, from acquisition to analyzing to ~~dis~~ use to disposal.

key components of good data governance $\rightarrow$ defining roles like data stewards, policies for security and compliance are enforced, ensuring data quality and maintaining metadata.

32 Bias AI Model

$\rightarrow$ refers to changes made from original data by humans and end up changing the original training data or AI algorithm leading to harsh and distorted outcomes and outputs.

that info is captured by machine learning or AI and becomes the data known and captured by machines

$\rightarrow$ In health care, data of women that was ~~was~~ underrepresented skew ~~per~~ predictive AI algorithms leading to results being less accurate for certain group of women compared to the other race.

33.

Data analyst	Data Scientist	Data Engineer	AI Engineer
Interprets data to provide actionable business insights	Predicts future trends and creates complex models	Builds robust data Pipelines and architecture	Build AI systems that make decisions and automatically
uses SQL, excel Power BI, Tableau	Python, R, SQL machine learning lib.	SQL, Spark, cloud Platforms, Kafka.	Integrates existing models into production
Structured Data cleans data	Structured & unstructured Data	structured and unstructured data	creates patterns
creates reports & graphs	Build ml models	works with cloud services	Build Products that think and act

34. Business Intelligence

- A set of technological processes for collecting, managing and analyzing organizational data to make decisions to take action.

→ Businesses prefer using BI tools because they speed up data analysis, which saves time, better understanding with customer behaviour and purchasing patterns, tracking of sales, marketing and financial performance. Tools used are Power BI, Tableau, Zoho analytics.

35. Section E : Personal Reflection

35. from question 12 until 15, It is fascinating how machines quickly replicate and mimic human behaviour and thinking ability. It is as almost in the future machines will erase the need of human intellect amongst us humans and will rely on machines even if information were to be bias, humans are easily manipulated as of course machines are.

36. Section C: APIs, Tools & developer Concepts, the entire section was a challenge even though I have compiled my findings about it I still have to go back and dwell much to it so I can be confident when tackling real life scenarios or having to apply actual coding, of course site like IBM, youtube videos from experts did help a lot.

37 They make tasks easier, save time and knowing that information is readily available at the tip of my fingers or just a voice recognition and it is impressive for example I can ask meta, GeminiAI or Siri for specific data and it will appear, ~~I have a fearless yet~~ I fear AI will replace a lot of human assistance as businesses will rely mostly on it but I believe that it makes life easier, like memories are kept in clouds and it is a pleasant thing or hospitals using AI for faster detection of diseases

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